

Types of Variables in Bash

1. Local Variables

- Available only in the current shell or function.

Example:

```
name="Alice"
echo $name
```

2. Environment Variables

- Global variables available to child processes.

Example:

```
export USERNAME="bob"
bash -c 'echo $USERNAME'
```

3. Shell Variables

- Built-in variables defined by the shell.

Examples: \$HOME, \$PWD, \$PATH, \$USER, \$SHELL, \$RANDOM

4. Positional Parameters

- Represent arguments passed to a script.

Example:

```
echo "First argument: $1"
```

5. Special Variables

- Provide information about shell or scripts.

Examples:

```
$0 (script name)
$? (exit status)
$$ (PID)
$@ (all arguments)
```

6. Array Variables

- Store multiple values.

Example:

```
fruits=("apple" "banana" "cherry")  
echo ${fruits[1]}
```

7. Read-only Variables

- Cannot be changed after assignment.

Example:

```
readonly pi=3.14
```